# SYNOPSIS

## Project Title

**Lumina: The Light Beyond Our Eyes**
*An AI-Powered Assistive System for Visually Impaired Individuals*

## Introduction

Lumina is an AI-powered assistive system designed to help visually impaired individuals navigate their surroundings more safely and independently. The system combines **Artificial Intelligence, Computer Vision, Optical Character Recognition (OCR), and Text-to-Speech (TTS)** technologies to provide real-time voice-based assistance.

## Problem Statement

Visually impaired individuals often face challenges while navigating unfamiliar surroundings, identifying obstacles, and accessing printed information such as signboards, notices, labels, and books. Many existing assistive technologies can be expensive or dependent on internet connectivity. Lumina aims to provide an affordable and accessible solution that converts visual information into meaningful audio instructions.

## Objective

The main objective of Lumina is to develop a simple and affordable AI-based assistive system that can:

- Detect common obstacles in the user's surroundings.
- Recognize printed text using OCR technology.
- Convert visual information into clear voice messages.
- Provide real-time environmental awareness.
- Improve safety, accessibility, and independent mobility.

## Proposed Solution

Lumina uses a camera to continuously capture the user's surroundings. The captured video is processed using **OpenCV**, while an object detection model such as **YOLOv8** identifies common objects and obstacles, including people, chairs, tables, and other objects.

At the same time, **EasyOCR or Tesseract OCR** recognizes printed text from signboards, notices, labels, books, and other surfaces. The detected objects and extracted text are then converted into spoken messages using **pyttsx3**, an offline Text-to-Speech library.

Through this process, Lumina enables users to receive important information about their surroundings through audio without relying on their vision.

## Working Principle

**Camera → Image Processing → Object Detection + OCR → Information Processing → Text-to-Speech → Voice Output**

1. The camera captures live images or video.
2. OpenCV processes the captured frames.
3. YOLOv8 detects objects and potential obstacles.
4. OCR identifies and extracts visible printed text.
5. The system processes the detected information.
6. pyttsx3 converts the information into speech.
7. The user receives the information through voice output.

## Technologies Used

| Technology | Purpose |
|---|---|
| **Python** | Primary programming language |
| **OpenCV** | Camera access and image/video processing |
| **YOLOv8** | Real-time object and obstacle detection |
| **EasyOCR / Tesseract OCR** | Printed text recognition |
| **pyttsx3** | Offline Text-to-Speech conversion |
| **Camera** | Capturing the surrounding environment |
| **Artificial Intelligence** | Visual information processing and interpretation |

## Key Features

- Real-time obstacle detection.
- Recognition of common surrounding objects.
- Printed text recognition.
- Voice-based environmental alerts.
- Offline Text-to-Speech functionality.
- Affordable and scalable design.
- Potential for wearable implementation.
- Supports greater independence for visually impaired users.

## Innovation and Uniqueness

The uniqueness of **Lumina** lies in combining **object detection, text recognition, and voice assistance** into a single assistive system. Unlike a system that only detects obstacles, Lumina can also identify and read useful printed information from the user's surroundings.

The use of offline technologies, particularly pyttsx3, provides the possibility of operating essential voice assistance without continuous internet connectivity, making the concept suitable for a low-cost and accessible solution.

## Benefits

Lumina can help visually impaired individuals to:

- Identify nearby obstacles.
- Navigate their surroundings more confidently.
- Read signboards, notices, labels, and other printed information.
- Access visual information independently.
- Reduce dependence on others for routine tasks.
- Improve safety and accessibility in everyday environments.

## Future Scope

The prototype can be further developed into wearable **AI-powered smart glasses** or a compact assistive device. Future enhancements may include:

- GPS-based outdoor navigation.
- Indoor navigation and location assistance.
- Face recognition with appropriate privacy safeguards.
- Currency and object recognition.
- Traffic signal and road-sign detection.
- Voice-controlled interaction.
- IoT integration.
- Emergency alerts and location sharing.
- Compact and lightweight wearable hardware.

## Conclusion

**Lumina: The Light Beyond Our Eyes** demonstrates how Artificial Intelligence can transform visual information into meaningful audio assistance. By integrating **computer vision, YOLOv8 object detection, OCR, and Text-to-Speech technology**, Lumina can provide visually impaired individuals with real-time information about obstacles and printed text.

The project represents a step toward creating affordable and accessible assistive technologies that can promote **safer navigation, greater independence, and improved access to information** for visually impaired individuals.